

ZIFENG WANG

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ABOUT ME

I am a Senior Research Scientist at Google Cloud AI Research, where I work on self-improving agents, environment scaling and synthetic data for post-training. My research has shipped in Gemini and multiple high-impact Google products. My PhD focused on continual learning, the problem of learning without forgetting.

EDUCATION

Northeastern University

Boston, MA

Ph.D. in Computer Engineering, GPA: 4.0/4.0

Sep 2018 – Aug 2023

- **Advisor:** Prof. Jennifer Dy
- **Thesis:** “Effective and Efficient Continual Learning”

Tsinghua University

Beijing, China

B.Eng. in Electronic Engineering, GPA: 92/100 (*top 5% out of 233*)

Aug 2014 – July 2018

WORK EXPERIENCE

Cloud AI Research, Google

Playa Vista, CA

Senior Research Scientist

July 2023 – Present

- Initiated and led multiple research projects on large language models (LLMs):
 - Self-improving agents and environment scaling [[EnvHarness](#), [RRSI](#), [SkillIOS \(NeurIPS'26\)](#), [ReasoningBank \(ICLR'26\)](#)]
 - Post-training and synthetic data [[Gemini-2.5 Tech Report](#), [Supervised RL \(ICLR'26\)](#), [Magnet \(ACL'25\)](#), [CodecLM \(NAACL'24\)](#)]
 - Automated multi-agent optimization [[Model Swarms \(ICML'25\)](#), [Heterogeneous Swarms \(NeurIPS'25\)](#)]
 - Efficient agents [[Budget-Aware \(COLM'26\)](#), [Speculative KD \(ICLR'25\)](#), [Model Routing \(ICLR'26\)](#)]
- Landed multiple research projects into high-impact Google products as research lead:
 - Gemini Model Series
 - Gemini Enterprise Agent Designer
 - Vertex AI Agent Engine Memory Bank
 - Proprietary self-improving agents for infrastructure
- Mentored 20+ research interns.
- Published 20+ papers at top-tier ML and NLP venues.

Google Research

Remote / Sunnyvale, CA

Research Intern; Advised by: Zizhao Zhang, Vincent Perot, Jacob Devlin

May 2022 – Jan 2023

- QueryForm: A zero-shot document entity extraction (DEE) Framework [[ACL'23](#)].
- Large-scale webpage-based pre-training for document understanding.

Google Research

Remote / Sunnyvale, CA

Research Intern; Advised by: Zizhao Zhang, Chen-Yu Lee, Han Zhang

June 2021 – May 2022

- Proposed the first prompting-based continual learning framework.
 - Learning to prompt for continual learning [[CVPR'22](#)].
 - DualPrompt [[ECCV'22](#)].

ACADEMIC EXPERIENCE

Machine Learning Group, Northeastern University

Boston, MA

Graduate Research Assistant; Advised by: Prof. Jennifer Dy

Sep 2018 – July 2023

- Effective and efficient continual learning [ICML'23, NeurIPS'22].
- Robust learning of neural networks [NeurIPS'21, ICDM'20].

Channing Laboratory, Harvard Medical School

Boston, MA

Student collaborator; Advised by: Prof. Peter J. Castaldi, Prof. Jennifer Dy

Sep 2018 – Sep 2021

- Smoking prediction via isoform-aware RNA-seq deep learning models [PLOS Computational Biology].

i-Vision Group, Tsinghua University

Beijing, China

Undergrad Research Assistant; Advised by: Prof. Jiwen Lu

Sep 2017 – Mar 2018

- Multi-object tracking via deep reinforcement learning [ECCV'18].

SELECTED PUBLICATIONS ([GOOGLE SCHOLAR](#))

(* indicates equal contribution; † indicates corresponding author or supervisor)

Preprints:

- [1] Chengsong Huang, **Zifeng Wang**[†], Rujun Han, Jun Yan, Yanfei Chen, Zoey CuiZhu, Ke Jiang, Peng Xia, Han Yu, Yufan Zhuang, Yifei Ming, Jiaqi Pan, Bhavana Dalvi Mishra, Jiaxin Huang, Burak Gokturk, Tomas Pfister, Chen-Yu Lee. “[EnvHarness: Awakening Static Worlds for Agent Learning](#)”. *ArXiv* 2026.
- [2] Peng Xia, Rujun Han, **Zifeng Wang**, Yanfei Chen, Yufan Zhuang, Yoonho Lee, Chengsong Huang, Han Yu, Zhongying CuiZhu, Yifei Ming, Huaxiu Yao, Burak Gokturk, Tomas Pfister, Chen-Yu Lee. “[RRSI: Regularized Recursive Self-Improvement of Agent Harnesses](#)”. *ArXiv* 2026.
- [3] **Zifeng Wang** and Gemini Team. “[Gemini 2.5: Pushing the frontier with advanced reasoning, multimodality, long context, and next generation agentic capabilities](#)”. *ArXiv* 2025.

Conference Papers:

- [4] Tengxiao Liu, **Zifeng Wang**[†], Jin Miao, I-Hung Hsu, Jun Yan, Jiefeng Chen, Rujun Han, Fangyuan Xu, Yanfei Chen, Ke Jiang, Samira Daruki, Yi Liang, William Yang Wang, Tomas Pfister, Chen-Yu Lee. “[Budget-Aware Tool-Use Enables Effective Agent Scaling](#)”. *COLM* 2026.
- [5] Siru Ouyang, Jun Yan, Yanfei Chen, Rujun Han, **Zifeng Wang**, Bhavana Dalvi Mishra, Rui Meng, Chun-Liang Li, Yizhu Jiao, Kaiwen Zha, Maohao Shen, Vishy Tirumalashetty, George Lee, Jiawei Han, Tomas Pfister, Chen-Yu Lee. “[SkillOS: Learning Skill Curation for Self-Evolving Agents](#)”. *NeurIPS* 2026.
- [6] Siru Ouyang, Jun Yan, I-Hung Hsu, Yanfei Chen, Ke Jiang, **Zifeng Wang**, Rujun Han, Long T. Le, Samira Daruki, Jiawei Han, Chen-Yu Lee, Tomas Pfister, *et al.* “[ReasoningBank: Scaling Agent Self-Evolving with Reasoning Memory](#)”. *ICLR* 2026.
- [7] Yihe Deng, I-Hung Hsu, Jun Yan, **Zifeng Wang**, Rujun Han, Gufeng Zhang, Yanfei Chen, Wei Wang, Tomas Pfister, Chen-Yu Lee. “[Supervised Reinforcement Learning: From Expert Trajectories to Step-wise Reasoning](#)”. *ICLR* 2026.
- [8] Wittawat Jitkittum, Harikrishna Narasimhan, Ankit Singh Rawat, Jeevesh Juneja, **Zifeng Wang**, Chen-Yu Lee, Pradeep Shenoy, Rina Panigrahy, Aditya Krishna Menon, Sanjiv Kumar. “[Universal Model Routing for Efficient LLM Inference](#)”. *ICLR* 2026.
- [9] Shangbin Feng, Wenxuan Ding, Alisa Liu, **Zifeng Wang**, Weijia Shi, Yike Wang, Zejiang Shen, Xiaochuang Han, Hunter Lang, Chen-Yu Lee, Tomas Pfister, Yejin Choi, Yulia Tsvetkov. “[When One LLM Drools, Multi-LLM Collaboration Rules](#)”. *ACL* 2026.
- [10] Shangbin Feng, **Zifeng Wang**[†], Palash Goyal, Yike Wang, Weijia Shi, Huang Xia, Hamid Palangi, Luke Zettlemoyer, Yulia Tsvetkov, Chen-Yu Lee, Tomas Pfister. “[Heterogeneous Swarms: Jointly Optimizing Model Roles and Weights for Multi-LLM Systems](#)”. *NeurIPS* 2025.

- [11] Shangbin Feng, **Zifeng Wang**[†], Yike Wang, Sayna Ebrahimi, Hamid Palangi, Lesly Miculicich, Achin Kuleshrestha, Nathalie Rauschmayr, Yejin Choi, Yulia Tsvetkov, Chen-Yu Lee, Tomas Pfister. “[Model Swarms: Collaborative Search to Adapt LLM Experts via Swarm Intelligence](#)”. *ICML 2025*.
- [12] Fan Yin, **Zifeng Wang**[†], I-Hung Hsu, Jun Yan, Ke Jiang, Yanfei Chen, Jindong Gu, Long T. Le, Kai-Wei Chang, Chen-Yu Lee, Hamid Palangi, Tomas Pfister. “[Magnet: Multi-turn Tool-use Data Synthesis and Distillation via Graph Translation](#)”. *ACL 2025*.
- [13] Zhen Tan, Jun Yan, I-Hung Hsu, Rujun Han, **Zifeng Wang**, Long T. Le, Yiwen Song, Yanfei Chen, Hamid Palangi, George Lee, Anand Iyer, Tianlong Chen, Huan Liu, Chen-Yu Lee, Tomas Pfister. “[In Prospect and Retrospect: Reflective Memory Management for Long-term Personalized Dialogue Agents](#)”. *ACL 2025*.
- [14] Justin Chih-Yao Chen, **Zifeng Wang**[†], Hamid Palangi, Rujun Han, Sayna Ebrahimi, Long T. Le, Vincent Perot, Swaroop Mishra, Mohit Bansal, Chen-Yu Lee, Tomas Pfister. “[Reverse Thinking Makes LLMs Stronger Reasoners](#)”. *NAACL 2025*.
- [15] Wenda Xu, Rujun Han, **Zifeng Wang**[†], Long T. Le, Dhruv Madeka, Lei Li, William Yang Wang, Rishabh Agarwal, Chen-Yu Lee, Tomas Pfister. “[Speculative Knowledge Distillation: Bridging the Teacher-Student Gap Through Interleaved Sampling](#)”. *ICLR 2025*.
- [16] Zilong Wang, **Zifeng Wang**[†], Long T. Le, Huaixiu Steven Zheng, Swaroop Mishra, Vincent Perot, Yuwei Zhang, Anush Mattapalli, Ankur Taly, Jingbo Shang, Chen-Yu Lee, Tomas Pfister. “[Speculative RAG: Enhancing Retrieval Augmented Generation through Drafting](#)”. *ICLR 2025*.
- [17] Lichang Chen, Hexiang Hu, Mingda Zhang, Yiwen Chen, **Zifeng Wang**, Yandong Li, Pranav Shyam, Tianyi Zhou, Heng Huang, Ming-Hsuan Yang, Boqing Gong. “[OmnixR: Evaluating Omni-modality Language Models on Reasoning across Modalities](#)”. *ICLR 2025*.
- [18] Zilong Wang, Hao Zhang, Chun-Liang Li, Julian Martin Eisenschlos, Vincent Perot, **Zifeng Wang**, Lesly Miculicich, Yasuhisa Fujii, Jingbo Shang, Chen-Yu Lee, Tomas Pfister. “[Chain-of-Table: Evolving Tables in the Reasoning Chain for Table Understanding](#)”. *ICLR 2024*.
- [19] **Zifeng Wang**, Chun-Liang Li, Vincent Perot, Long T. Le, Jin Miao, Zizhao Zhang, Chen-Yu Lee, Tomas Pfister. “[CodecLM: Aligning Language Models with Tailored Synthetic Data](#)”. *Findings of NAACL 2024*.
- [20] **Zifeng Wang**, Zizhao Zhang, Jacob Devlin, Chen-Yu Lee, Guolong Su, Hao Zhang, Jennifer Dy, Vincent Perot, Tomas Pfister. “[QueryForm: A Simple Zero-shot Form Entity Query Framework](#)”. *Findings of ACL 2023*.
- [21] **Zifeng Wang**, Zheng Zhan, Yifan Gong, Yucai Shao, Stratis Ioannidis, Yanzhi Wang, Jennifer Dy. “[DualHSIC: HSIC-Bottleneck and Alignment for Continual Learning](#)”. *ICML 2023*.
- [22] **Zifeng Wang**^{*}, Zheng Zhan^{*}, Yifan Gong, Geng Yuan, Wei Niu, Tong Jian, Bin Ren, Stratis Ioannidis, Yanzhi Wang, Jennifer Dy. “[SparCL: Sparse Continual Learning on the Edge](#)”. *NeurIPS 2022*.
- [23] **Zifeng Wang**, Zizhao Zhang, Sayna Ebrahimi, Ruoxi Sun, Han Zhang, Chen-Yu Lee, Xiaoqi Ren, Guolong Su, Vincent Perot, Jennifer Dy, Tomas Pfister. “[DualPrompt: Complementary Prompting for Rehearsal-free Continual Learning](#)”. *ECCV 2022*.
- [24] **Zifeng Wang**, Zizhao Zhang, Chen-Yu Lee, Han Zhang, Ruoxi Sun, Xiaoqi Ren, Guolong Su, Vincent Perot, Jennifer Dy, Tomas Pfister. “[Learning to Prompt for Continual Learning](#)”. *CVPR 2022*.
- [25] Tong Jian^{*}, **Zifeng Wang**^{*}, Yanzhi Wang, Jennifer Dy, Stratis Ioannidis. “[Pruning Adversarially Robust Neural Networks without Adversarial Examples](#)”. *ICDM 2022*.
- [26] **Zifeng Wang**^{*}, Tong Jian^{*}, Aria Masoomi, Stratis Ioannidis, Jennifer Dy. “[Revisiting Hilbert-Schmidt Information Bottleneck for Adversarial Robustness](#)”. *NeurIPS 2021*.
- [27] **Zifeng Wang**^{*}, Tong Jian^{*}, Kaushik Chowdhury, Yanzhi Wang, Jennifer Dy, Stratis Ioannidis. “[Learn-Prune-Share for Lifelong Learning](#)”. *ICDM 2020*.
- [28] **Zifeng Wang**, Batool Salehi, Andrey Gritsenko, Kaushik Chowdhury, Stratis Ioannidis, Jennifer Dy. “[Open-World Class Discovery with Kernel Networks](#)”. *ICDM 2020*. **Best Paper Candidate**.
- [29] Aria Masoomi, Chieh Wu, Tingting Zhao, **Zifeng Wang**, Peter Castaldi, Jennifer Dy. “[Instance-wise Feature Grouping](#)”. *NeurIPS 2020*.
- [30] Andrey Gritsenko^{*}, **Zifeng Wang**^{*}, Jennifer Dy, Kaushik Chowdhury, Stratis Ioannidis. “[Finding a ‘New’ Needle in the Haystack: Unseen Radio Detection in Large Populations Using Deep Learning](#)”. *DySPAN 2019*, **Best Paper Award**.
- [31] Liangliang Ren, Jiwen Lu, **Zifeng Wang**, Qi Tian, Jie Zhou. “[Collaborative Deep Reinforcement Learning for Multi-Object Tracking](#)”. *ECCV 2018*.

Journal Papers:

- [32] Haizhou Shi, Zihao Xu, Hengyi Wang, Weiyi Qin, Wenyuan Wang, Yibin Wang, **Zifeng Wang**, Sayna Ebrahimi, Hao Wang. “[Continual Learning of Large Language Models: A Comprehensive Survey](#)”. *ACM Computing Surveys* (2025).
- [33] Ruoxi Sun, Serkan Ö. Arik, Alex Muzio, Lesly Miculicich, Satya Gundabathula, Pengcheng Yin, Hanjun Dai, Hootan Nakhost, Rajarishi Sinha, **Zifeng Wang**, Tomas Pfister. “[SQL-PaLM: Improved Large Language Model Adaptation for Text-to-SQL](#)”. *Transactions on Machine Learning Research* (2024).
- [34] **Zifeng Wang**, Aria Masoomi, Zhonghui Xu, Adel Boueiz, Sool Lee, Tingting Zhao, Russell Bowler, Michael Cho, Edwin K. Silverman, Craig Hersch, Jennifer Dy, Peter J. Castaldi. “[Improved Prediction of Smoking Status via Isoform-Aware RNAseq Deep Learning Models](#)”. *PLOS Computational Biology* 17(10): e1009433.
- [35] Tingting Zhao*, **Zifeng Wang***, Aria Masoomi, Jennifer Dy. “[Deep Bayesian Unsupervised Lifelong Learning](#)”. *Neural Networks* 149, 95–106.
- [36] Tong Jian, Yifan Gong, Zheng Zhan, Runbin Shi, Nasim Soltani, **Zifeng Wang**, Jennifer Dy, Kaushik Chowdhury, Yanzhi Wang, Stratis Ioannidis. “[Radio Frequency Fingerprinting on the Edge](#)”. *IEEE Transactions on Mobile Computing* 21(11), 4078–4093.

HONORS AND AWARDS

Outstanding Graduate Student in Research , Northeastern University, 2023	Boston, MA
Scholar Award , NeurIPS 2022	New Orleans, LA
Best Paper Candidate , ICDM 2020	Virtual
Best Paper Award , IEEE DySPAN 2019	Newark, NJ
Travel Award , NeurIPS 2019	Vancouver, Canada
Dean’s Fellowship , Northeastern University, 2018	Boston, MA
Outstanding Undergraduate Scholarship , Tsinghua University, 2016	Beijing, China

ACADEMIC SERVICES

Area Chair: ACL ARR
Conference Reviewer: NeurIPS, ICML, ICLR, CVPR, ICCV
PC Member: SDM
Journal Reviewer: TPAMI, TMLR, Neural Networks

INVITED TALKS

EnvHarness: Awakening Static Worlds for Agent Learning , ServiceNow Research and Mila	Sep 2026
Automated Multi-Agent System Optimization , Google DeepMind	Aug 2025
CodecLM: Synthetic Data Generation for Large Language Models , Google Cloud	Oct 2024
Effective and Efficient Continual Learning , CFAR, A*STAR	July 2023
SparCL: Sparse Continual Learning on the Edge , ContinualAI	Feb 2023
QueryForm: A Simple Zero-shot Form Entity Query Framework , Google Cloud	Nov 2022
Revisiting HSIC Bottleneck for Adversarial Robustness , INFORMS; AI Times	June – Oct 2022
Prompting-based Continual Learning , ContinualAI; Google Cloud; NTU AI Talks	Mar – Sep 2022